



## MA 570: Data-driven Dynamical Systems

|              |                                   |
|--------------|-----------------------------------|
| Course Code  | MA 570                            |
| Course Name  | Data-driven Dynamical Systems     |
| L-T-P-C      | 2.5-0-0.5-3                       |
| Intended for | M.Sc. / BTech / MTech/ PhD (SMSS) |
| Approval     | 54th BoA (May 2024)               |

### Course Contents

- Dimensionality reduction and transformations: Pseudo-inverse, least-squares, regression, singular value decomposition (SVD), principal component analysis (PCA), Discrete Fourier Transform, Fast Fourier Transform, Transforming Partial differential equations. (8 hours)
- Basics of machine learning: Basic definitions, Types of learning: Supervised learning and Unsupervised learning, Linear regression, Nonlinear Regression and Gradient Descent, Over and Under- Determined Systems, Least-Squares Fitting Methods, The Pareto Front, Model Selection: Cross-Validation, Model Selection: Information Criteria. (8 hours)
- Basics of neural networks: Perceptron, single-layer neural networks, multilayer neural networks and Activation Functions, Backpropagation, Neural networks for Dynamical Systems – Recurrent Neural Networks, Generative Adversarial Networks (GANs). (8 hours)
- Data Driven Dynamical Systems: Overview, motivations, and challenges, Dynamic mode decomposition (DMD), Sparse identification of nonlinear dynamics (SINDy), Koopman operator theory, Data-driven Koopman analysis, Model Reduction and System Identification. (11 hours)

### Lab Components

Singular value decomposition (SVD), principal component analysis (PCA), Linear regression, Least-Squares Fitting Methods, Single-layer linear neural network, neural networks for dynamical systems, DMD, Sparse identification of nonlinear dynamics and their variants. (7 hours)

### Text Books

1. Strang, Gilbert, **Linear algebra and learning from data**, Vol. 4, Wellesley- Cambridge Press, 2019.
2. C.M. Bishop, **Pattern Recognition and Machine Learning**, Springer, 2006.
3. Ian Goodfellow, Yoshua Bengio and Aaron Courville, **Deep learning**, MIT Press, 2016.
4. Vega, J.M. and Le Clainche, S., **Higher Order Dynamic Mode Decomposition and its Applications**, Academic Press, 2020.

### References

1. Tom Mitchell, **Machine Learning**, Mc-Graw Hills, 1997.
2. Mauroy, A., Susuki, Y. and Meizic., **Koopman Operator in Systems and Control**, Springer, 2020